# Java Interview Coding Assignment

**Collections & Java 8 Features — 50 Questions**

## Instructions

• Write complete executable Java programs for each question.
• Explain your approach and important Java concepts used.
• Focus on clean, readable, interview-ready code.

## Part A — Collections (1–25)

**1. Remove duplicates from an ArrayList**
Write a Java program to remove duplicate elements from an ArrayList without using a Set.

**2. Find duplicate elements**
Given an ArrayList, find all duplicate elements.

**3. Find frequency of elements**
Find the frequency of each element in an ArrayList.

**4. Find the second largest number**
Find the second largest number from a List.

**5. Find the second smallest number**
Find the second smallest number from a List.

**6. Sort an ArrayList**
Sort an ArrayList in ascending and descending order.

**7. Reverse an ArrayList**
Reverse an ArrayList without using Collections.reverse().

**8. Find common elements**
Find common elements between two ArrayList objects.

**9. Find missing elements**
Given two lists, find the elements present in the first list but missing from the second.

**10. Move zeros to the end**
Move all 0 values in an ArrayList to the end while maintaining the order of other elements.

**11. Count even and odd numbers**
Count the number of even and odd elements in a list.

**12. Find maximum and minimum**
Find the maximum and minimum values from a list without using Collections.max() or Collections.min().

**13. Remove a specific element**
Remove all occurrences of a given number from an ArrayList.

**14. Merge two lists**
Merge two ArrayList objects and remove duplicates.

**15. Convert List to Set**
Convert an ArrayList into a HashSet and explain what happens to duplicate elements.

**16. Find first non-repeated element**
Given a list of integers, find the first element that occurs only once.

**17. Find first repeated element**
Find the first repeated element in a list.

**18. Sort employees by salary**
Create an Employee class with id, name, and salary. Sort employees based on salary.

**19. Sort employees by name**
Sort a list of employees alphabetically by employee name.

**20. Sort employees using multiple conditions**
Sort employees by salary. If salaries are equal, sort by name.

**21. HashMap character frequency**
Find the frequency of each character in a given String using HashMap.

**22. HashMap word frequency**
Find the frequency of each word in a sentence.

**23. Find duplicate words**
Find all duplicate words from a given sentence using HashMap.

**24. Find employee with highest salary**
Store employees in a HashMap and find the employee with the highest salary.

**25. Group employees by department**
Given employees with id, name, salary, and department, group employees based on department.

## Part B — Java 8 Features (26–50)

**26. Print list using Lambda**
Print all elements of a List using a Lambda expression.

**27. Filter even numbers**
Use Java 8 Lambda and Stream API to print all even numbers from a list.

**28. Filter numbers greater than 50**
Find all numbers greater than 50 using Lambda and Stream.

**29. Sort using Lambda**
Sort a list of integers in ascending order using Lambda.

**30. Sort strings by length**
Sort a list of Strings based on their length using Lambda.

**31. Find sum of numbers**
Find the sum of all elements in a list using Stream API.

**32. Find average**
Find the average of numbers in a list using Stream API.

**33. Find maximum and minimum**
Find maximum and minimum numbers using Stream API.

**34. Remove duplicates using Stream**
Remove duplicate elements from a list using distinct().

**35. Find duplicate elements using Stream**
Find duplicate elements from a list using Stream API.

**36. Count even numbers**
Count how many even numbers exist in a list using Stream API.

**37. Find second highest number**
Find the second highest number from a list using Stream API.

**38. Find second lowest number**
Find the second lowest number using Stream API.

**39. Convert strings to uppercase**
Convert all strings in a list to uppercase using Stream API.

**40. Filter strings starting with A**
Find all strings starting with "A" using Stream API.

**41. Employee highest salary**
Given a list of employees, find the employee with the highest salary using Stream API.

**42. Employee second highest salary**
Find the employee having the second-highest salary.

**43. Employees with salary > 50,000**
Find all employees whose salary is greater than ■50,000.

**44. Group employees by department**
Use Collectors.groupingBy() to group employees based on department.

**45. Count employees by department**
Find the number of employees in each department using groupingBy() and counting().

**46. Average salary by department**
Find the average salary of employees in each department.

**47. Highest-paid employee in each department**
Find the employee with the highest salary from every department.

**48. Convert List to Map**
Convert a list of employees into a Map using Collectors.toMap().

**49. Partition employees by salary**
Divide employees into two groups: salary >= 50,000 and salary < 50,000. Use Collectors.partitioningBy().

**50. Find duplicate employees based on name**
Given a list of employees, find duplicate employee names using Java 8 Streams and Collections.